

The Uncertainty Principle in Time-Series Momentum

A signal-processing bound on lag, turnover, and regime detection

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May 2026

Abstract

Momentum strategies filter historical return series to isolate persistent low-frequency trends from high-frequency noise [1, 2]. The Gabor–Heisenberg uncertainty principle [3] sets a lower bound on the simultaneous localisation of a signal in time and frequency. This note maps the spectral spread σ_ξ and temporal spread σ_t of a linear filter to three operational floors: signal lag, expected turnover, and regime-detection delay. These floors are mathematical consequences of the inequality $\sigma_t \sigma_\xi \geq 1/(4\pi)$ and cannot be removed by adaptive methods or higher-frequency data. The bounds are structural; they do not by themselves limit profitability, but they constrain the speed-accuracy trade-off that any trend-following strategy must accept.

1 Introduction

Time-series momentum (TSMOM) strategies use an asset’s own historical returns to determine its position [1]. A TSMOM indicator is a linear, time-invariant (LTI) filter applied to a continuous-time return process. We show that such a filter is governed by the Gabor–Heisenberg uncertainty principle [3]. This principle imposes a hard limit on how precisely the filter can be localised simultaneously in time and frequency. The limit is translated into quantitative lower bounds on the filter’s lag, turnover, and the minimum delay required to detect a regime change.

Moskowitz, Ooi, and Pedersen [1] document that time-series momentum yields significant returns across a broad universe of asset classes and horizons, with lookback windows typically ranging from one to twelve months. Lesmond, Schill, and Zhou [4] challenge the net profitability of momentum strategies, showing that the transaction costs required to implement them consume most of the gross return. The bounds derived here offer a structural explanation for that finding: the same uncertainty principle that makes a trend detectable also sets a floor on the turnover required to trade it, and therefore on the minimum implementable cost. Jegadeesh and Titman [2] establish the cross-sectional counterpart of TSMOM; the filter-theoretic framework developed here applies equally to that setting, as discussed in Section 8. For century-long empirical evidence that trend-following premia persist across regimes, see Hurst, Ooi, and Pedersen [5].

2 The Gabor–Heisenberg Uncertainty Principle

Let $g(u) \in L^2(\mathbb{R})$ be a window function of unit energy, $\|g\|_2 = 1$ [3, 6]. The temporal centre μ_t and temporal variance σ_t^2 are defined through the energy density $|g(u)|^2$:

$$\mu_t = \int u |g(u)|^2 du, \tag{1}$$

$$\sigma_t^2 = \int (u - \mu_t)^2 |g(u)|^2 du. \quad (2)$$

The Fourier transform $\hat{g}(\xi) = \int g(u) e^{-2\pi i \xi u} du$ preserves the norm, $\|\hat{g}\|_2 = 1$; we analogously define the spectral centre and spectral variance:

$$\mu_\xi = \int \xi |\hat{g}(\xi)|^2 d\xi, \quad (3)$$

$$\sigma_\xi^2 = \int (\xi - \mu_\xi)^2 |\hat{g}(\xi)|^2 d\xi. \quad (4)$$

The Gabor–Heisenberg inequality states [3]

$$\sigma_t \cdot \sigma_\xi \geq \frac{1}{4\pi} \quad (5)$$

with equality if and only if g is a Gabor atom (a Gaussian-modulated sinusoid). The inequality sets the minimum possible area of a resolution cell in the time-frequency plane. For a low-pass filter centred at zero frequency, $\mu_\xi = 0$, then σ_ξ is the root-mean-square width of the passband.

3 Momentum as a Linear Filter

A discrete TSMOM indicator is a weighted sum of past log-returns $\{r_t\}$ [1, 2]:

$$m_t = \sum_{k=0}^{\infty} w_k r_{t-k}. \quad (6)$$

Common weight sequences are uniform over a window of length W (simple moving average) or geometric (exponential moving average):

$$\begin{aligned} \text{SMA: } w_k &= \frac{1}{W} \quad \text{for } k < W. \\ \text{EMA: } w_k &= \alpha(1 - \alpha)^k, \quad \alpha \in (0, 1). \end{aligned}$$

To connect with the continuous-time uncertainty principle [3, 6], we model the return series as a continuous signal $r(t)$ and replace the weight sequence by a continuous window function $g(u)$ of the same shape. The discrete filter is the sampled version of the continuous convolution

$$m(t) = (r * g)(t) = \int_{-\infty}^{\infty} r(\tau) g(t - \tau) d\tau \quad (7)$$

where $g(u)$ is causal (non-zero only for $u \in [0, W]$) and satisfies $\|g\|_2 = 1$. The fundamental scaling of temporal and spectral spreads carries over to the sampled case; only minor constant adjustments arise from finite sampling.

The rectangular window (SMA analog), the Gaussian window, and the exponential window are the most common examples. Their precise definitions and temporal spreads are given in Table 1. For all these shapes, the temporal spread σ_t is proportional to the lookback length W :

$$\sigma_t = cW, \quad (8)$$

where the constant c depends on the window shape.

Note: Why $\sigma_t \propto W$?

The proportionality follows directly from the scaling of each window's support. For the rectangular window, $|g(u)|^2$ is a uniform density on $[0, W]$; its variance is $W^2/12$, so $\sigma_t = W/\sqrt{12}$. The Gaussian window has $|g(u)|^2$ exactly equal to a zero-mean Gaussian with variance σ^2 ; defining the effective lookback as the span $W = 4\sigma$ (which contains $\approx 95\%$ of the energy) gives $\sigma_t = W/4$. For the exponential window, $|g(u)|^2$ is an exponential density with variance $\tau^2/2$; identifying the lookback with the time constant τ yields $\sigma_t = \tau/\sqrt{2}$. In all cases, stretching the window by a factor a multiplies σ_t by the same factor, establishing the linear relation.

Table 1: Window functions, their temporal spreads, and the uncertainty premium.

Window	$g(u)$ (normalised)	σ_t	$\sigma_t \sigma_\xi$
Rectangular (SMA)	$\frac{1}{\sqrt{W}} \mathbf{1}_{[0, W]}(u)$	$\frac{W}{\sqrt{12}}$	$> 1/(4\pi)$
Gaussian	$\frac{1}{(2\pi\sigma^2)^{1/4}} e^{-u^2/(4\sigma^2)} \quad (W \approx 4\sigma)$	$\frac{W}{4}$	$= 1/(4\pi)$
Exponential (EMA)	$\sqrt{\frac{2}{\tau}} e^{-u/\tau} \mathbf{1}_{[0, \infty)}(u)$	$\frac{\tau}{\sqrt{2}}$	$> 1/(4\pi)$

Every non-Gaussian window pays an uncertainty premium: for the same σ_t , its spectral spread σ_ξ is strictly larger than the Gabor minimum.

4 The Resolution Cell of a Momentum Filter

The time-frequency resolution of a momentum filter is most naturally described using the Short-Time Fourier Transform (STFT), a framework developed in detail by Mallat [6]. The STFT of $r(t)$ with respect to a window g is

$$\mathcal{V}_g r(\tau, \xi) = \int_{-\infty}^{\infty} r(t) g(t - \tau) e^{-2\pi i \xi t} dt. \quad (9)$$

Its squared magnitude $S(\tau, \xi) = |\mathcal{V}_g r(\tau, \xi)|^2$ (the spectrogram) is the energy density of the signal in the time-frequency plane. A low-pass momentum filter averages the spectrogram over a frequency band; its output at time t is effectively the zero-frequency component of the STFT plus a remainder that depends on the filter shape. Because the window has finite temporal spread σ_t , the STFT cannot localise energy to a point. Every measurement at (τ, ξ) is a weighted average over a resolution cell whose dimensions we now identify.

Note: Resolution cell dimensions.

Since σ_t and σ_ξ are the RMS spreads of the window g in time and frequency respectively, they directly measure the extent of the averaging region around any point (τ, ξ) in the time-frequency plane. A window concentrated within $\pm\sigma_t$ of its centre in time cannot distinguish events separated by less than σ_t , and analogously in frequency. The dimensions of the resolution cell are therefore

$$\Delta t_{res} = \sigma_t, \quad \Delta \xi_{res} = \sigma_\xi.$$

The Gabor–Heisenberg inequality bounds the cell area [3]:

$$\Delta t_{res} \cdot \Delta \xi_{res} \geq \frac{1}{4\pi}. \quad (10)$$

If the true trend has a characteristic frequency ξ_0 , the filter output at time is contaminated by all fluctuations inside the band $[\xi_0 - \sigma_\xi/2, \xi_0 + \sigma_\xi/2]$. A smaller σ_t (faster reaction) forces a larger σ_ξ (more noise); a smaller σ_ξ (cleaner trend isolation) forces a larger σ_t (more lag).

5 Signal Lag

5.1 Group delay of a linear-phase window

Consider a symmetric window centred at $W/2$ (linear-phase). Its group delay for a slowly varying component equals $W/2$. With $\sigma_t = cW$ the lag is

$$\text{Lag} = \frac{W}{2} = \frac{1}{2c}\sigma_t. \quad (11)$$

5.2 Lower bound from the resolution requirement

Suppose the strategy must isolate a trend near frequency ξ_0 while rejecting fluctuations within a bandwidth $\Delta\xi$. This requires $\sigma_\xi \leq \Delta\xi$. Applying the Gabor bound $\sigma_\xi \geq 1/(4\pi\sigma_t)$ gives $\sigma_t \geq 1/(4\pi\Delta\xi)$, hence

$$\text{Lag} \geq \frac{1}{8\pi c\Delta\xi}. \quad (12)$$

5.3 Examples

For a rectangular window, $c = 1/\sqrt{12}$, yielding $\text{Lag} \geq 0.14/\Delta\xi$; for the Gaussian, $c = 1/4$, yielding $\text{Lag} \geq 0.16/\Delta\xi$.

If the target is a monthly trend ($\xi_0 \approx 1/21$ days⁻¹) and the filter must suppress fluctuations shorter than 10 days ($\Delta\xi \approx 0.052$ days⁻¹), the bound yields a minimum lag of roughly 3 days. If one must also distinguish a 21-day trend from a 25-day reversal ($\Delta\xi \approx 0.0076$ days⁻¹), the minimum lag jumps to about 21 days. A standard 252-day momentum signal [1] targeting that band therefore carries an irreducible lag of approximately 126 days (since $\text{Lag} \approx W/2$, as derived in Section 5.1).

6 A Lower Bound on Turnover

Turnover is the expected rate at which the filter output $m(t)$ changes sign, triggering a position flip.

6.1 Rice’s formula

Assume that returns are locally white noise with a flat spectrum $S_r(\xi) = \text{const}$. Rice [7], in his classical treatment of random noise in electrical circuits, derives a formula for the expected rate at which a stationary Gaussian process crosses zero. We apply it here to the filtered signal $m(t)$; this yields a clean analytic expression for turnover, and the proportionality between crossing rate and spectral width holds more generally for any stationary process with a continuous spectrum.

Note: Gaussian assumption.

The assumption of Gaussian returns is a modelling convenience for Rice's formula. The essential proportionality between turnover and σ_ξ (the spectral width) depends only on the second-order structure of the process and remains valid for any stationary, zero-mean process with a continuous spectrum, provided one interprets the crossing rate as the expected number of zero crossings of a Gaussian process with the same spectrum.

Under these assumptions, the filtered signal $m(t)$ is a stationary zero-mean Gaussian process with power spectrum

$$S_m(\xi) = |\hat{g}(\xi)|^2 S_r(\xi) \propto |\hat{g}(\xi)|^2. \quad (13)$$

Rice's formula [7] gives the expected zero-crossing rate

$$\rho_0 = \frac{1}{\pi} \sqrt{\frac{\int \xi^2 S_m(\xi) d\xi}{\int S_m(\xi) d\xi}} = \frac{1}{\pi} \sigma_\xi, \quad (14)$$

because the ratio equals the second spectral moment, which is σ_ξ^2 for a symmetric low-pass filter centred at zero frequency.

6.2 Turnover bound

Substituting the Gabor bound $\sigma_\xi \geq 1/(4\pi\sigma_t)$ and $\sigma_t = cW$ [3],

$$\mathbb{E}[\text{turnover}] \geq \frac{1}{4\pi^2 cW} \approx \begin{cases} 0.088/W & (\text{rectangular}), \\ 0.101/W & (\text{Gaussian}). \end{cases} \quad (15)$$

Halving the lookback at least doubles the minimum expected turnover. Transaction costs are proportional to turnover [4], so the uncertainty principle imposes a floor on the implementation shortfall that cannot be removed by execution algorithms alone. This connects directly to the empirical finding of Lesmond, Schill, and Zhou [4]: if gross momentum returns are approximately proportional to lookback length, as the empirical evidence in Moskowitz et al. roughly suggests [1], while minimum turnover scales as $1/W$, transaction costs erode a fixed fraction of returns regardless of the horizon chosen. The uncertainty principle thus provides a structural underpinning for their conclusion that momentum profits are largely illusory net of costs.

7 Regime Changes and Detection Delay

A regime change is a shift in the dominant frequency of the return process from ξ_{old} to ξ_{new} , with gap $\Delta\xi_{\text{regime}} = |\xi_{\text{new}} - \xi_{\text{old}}|$. To distinguish the two frequencies, the filter's spectral spread must be smaller than the gap: $\sigma_\xi \leq \Delta\xi_{\text{regime}}$. Combining this with $\sigma_\xi \geq 1/(4\pi\sigma_t)$ and $\sigma_t = cW$ yields

$$W \geq \frac{1}{c \cdot 4\pi \cdot \Delta\xi_{\text{regime}}}. \quad (16)$$

A shorter window cannot resolve the transition; it sees a single blurred component. The detection delay (the time after the shift until the filter output fully reflects the new regime) is therefore at least of order $1/\Delta\xi_{\text{regime}}$.

A direct consequence: a momentum strategy cannot simultaneously stay positioned for the old regime and act on the new one [1, 5]. The drawdown at a reversal is not a modelling failure but a necessary byproduct of the spectral resolution that identified the preceding trend.

8 Design Implications

Selecting a lookback W and a window shape fixes a point (σ_t, σ_ξ) in the feasible region $\sigma_t \sigma_\xi \geq 1/(4\pi)$. For the Gaussian window the point lies on the boundary $\sigma_t \sigma_\xi = 1/(4\pi)$; for other windows it lies strictly above it. That choice jointly determines the strategy’s minimum lag, minimum turnover, and regime-detection delay. From the bounds derived above, the following principles emerge:

1. **Match lookback to target frequency.** The resolution cell must cover the frequency band of the trend [1, 5]. Choose $W \approx k/\xi_{\text{target}}$ with $k \geq 2$ so that the cell is centred on the signal. A mismatched window places the cell outside the band and wastes resolution on noise.
2. **Budget for structural lag.** The $\approx W/2$ group delay is a direct consequence of the window length. Execution systems should account for it; the delay cannot be removed by faster data or smarter execution.
3. **Diversify across lookbacks.** A portfolio of signals with different lookbacks $\{W_1, W_2, \dots\}$ places several resolution cells in the time-frequency plane [5]. The correlation between two such signals is bounded from below by the overlap integral of their resolution cells, a quantity that can be computed from the window functions. This provides a rigorous limit on the diversification benefit achievable by combining momentum horizons.
4. **Prefer the Gaussian window.** The Gaussian uniquely attains the lower bound $\sigma_t \sigma_\xi = 1/(4\pi)$ [3]. Any other shape sacrifices more frequency selectivity for the same temporal resolution, or more lag for the same noise rejection.
5. **Handle universe heterogeneity in cross-sectional settings.** When applying a momentum filter to a portfolio of assets [2], a uniform lookback will be poorly matched to assets with different intrinsic trend frequencies. Stratifying lookbacks by asset class aligns each filter’s resolution cell with the local trend band, avoiding the noise that a mismatched window introduces into the cross-sectional ranking.

9 Limitations and What This Paper Does Not Claim

The bounds derived here apply to linear, time-invariant filters. Non-linear or adaptive filters may exhibit different trade-offs, though the resolution cell concept can often be extended by local linearisation.

The analysis assumes that the return process is locally stationary, i.e., that a single dominant frequency persists over the lookback window. This is an instance of the broader assumption of local translation symmetry, a standard condition in time-frequency analysis under which the STFT provides a consistent spectral representation [6]. When this symmetry breaks (e.g., during sharp structural breaks), the filter output loses its interpretation as a trend estimate, and the bounds may not describe the transient behaviour before a new local symmetry is established.

The floors on lag and turnover are lower bounds on those costs; they do not place an upper bound on the strategy’s Sharpe ratio or profitability [4, 5]. A strategy could operate

above these floors and still perform well, or it could respect the floors and perform poorly for other reasons (e.g., poor volatility scaling). The results do not imply that momentum returns are entirely explained by a delay premium; only that part of the return may be understood as compensation for bearing an irreducible timing risk, a hypothesis that requires separate empirical testing.

10 Future Work

The most immediate extension is to relax the assumption of local translation symmetry. In a non-stationary environment the window function itself must adapt, leading to time-varying temporal and spectral spreads that trace a path in the (σ_t, σ_ξ) plane. The geometry of such paths and their consequences for regime detection and turnover are the subject of ongoing work.

11 Conclusion

The Gabor–Heisenberg inequality $\sigma_t \sigma_\xi \geq 1/(4\pi)$ is a hard mathematical limit on every linear momentum filter [3]. It translates into irreducible lower bounds on signal lag, turnover [4], and regime-detection delay. No adaptive windowing, multi-scale decomposition, or sampling refinement can shrink the fundamental area of the resolution cell. Acknowledging this limit provides a rigorous basis for window selection, realistic capacity planning, and a clearer separation of structural costs from alpha in trend-following strategies [1, 5].

References

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